

Maximum Likelihood Estimator. Suppose we have iid samples $x_1, x_2, \dots, x_N$ from pdf or pmf $f(x; \theta)$. Then the joint density/probability is $f(x; \theta) = \prod_{i=1}^N f(x_i; \theta)$ and $\log L(\theta) = \sum_{i=1}^N \log [f(x_i; \theta)]$. The ML estimator $\hat{\theta}$ of θ is $\arg \max \log L(\theta)$.